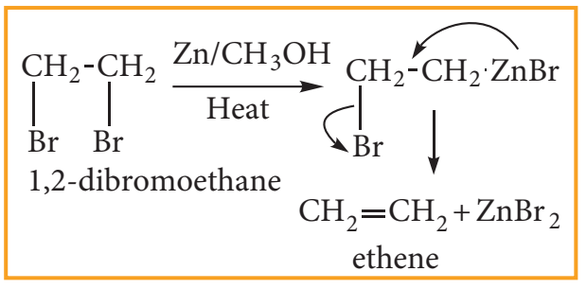



### (4) Preparation of alkenes from vicinal dihalogen derivative of alkanes or vicinal dihalides

The compound in which two halogen atoms are attached to adjacent carbon-atoms are called as vicinal dihalides. When vicinal dihalides are warmed with granulated zinc in methanol, they lose a molecule of $ZnX_2$ to form an alkene.

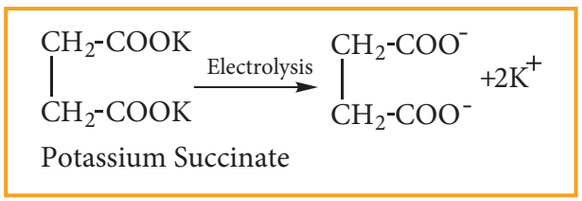



**Evaluate Yourself**

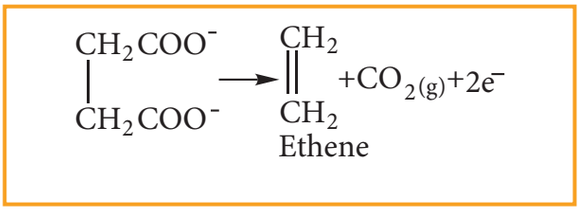
14) How propene is prepared form 1, 2-dichloro propane?

### (5) Preparation of ethene by kolbe's electrolytic method:

When an aqueous solution of potassium succinate is electrolyzed between two platinum electrodes, ethene is produced at the anode.




At anode




### 13.3.2. Physical properities of alkenes:

The first three members (Ethene, Propene and Butene) are gases, next fourteen members are liquids and the higher alkenes are waxy solids. They are all colourless and odourless except ethene which has a sweet smell.

1. The melting and boiling point of alkenes increases along the homologous series. Like alkanes, straight chain alkenes have high boiling point compared to its isomeric branched alkenes.

2. Alkenes are slightly soluble in water but readily in organic solvents.

### 13.3.3. Chemical properties of alkenes:

Alkenes are more reactive than alkanes due to the presence of a double bond. The σ- bond is strong but the π- bond is weak. The typical reactions of alkenes involve addition of an electrophile across the double bonds proceeding through ionic mechanism. However addition reactions proceed through free-radical mechanism also. Ozonolysis and polymerization are some of the characteristic reactions of alkenes.

### (i) Addition Reactions

### (ii) Addition of hydrogen: (Hydrogenation of alkenes)

hydrogen adds on to alkenes in the presence of a metal catalyst (Ni, Pd (or) Pt) to yield corresponding alkanes. This is known



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